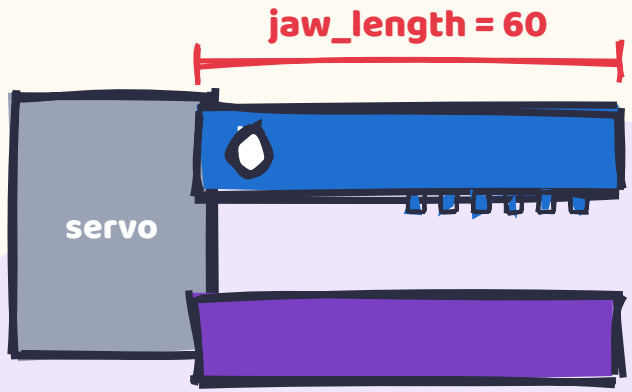
Tech Leaders · Class 15

Parametric Robot Gripper Design

Design once, resize forever

Parameters | Constraints | Onshape / OpenSCAD | Print & test



Did you know?
Car makers design one parametric seat frame, then change a few numbers to fit dozens of different car models.

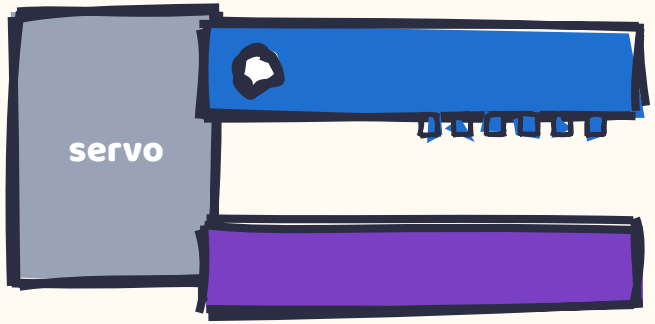
LEARN

Let's think...

One gripper for an egg AND a bottle?

Different jobs

A factory robot must pick up eggs today and juice bottles tomorrow. Redrawing the gripper from scratch each time takes days. What if the design were built from **numbers**, so changing 'object_width = 45' to '70' redesigned the whole thing in seconds?



Buy a new robot?

Redraw it?

Change a number!

Today: you design a **parametric gripper finger**, make two variants for different objects, print and test them.

LEARN

Learning objectives

By the end of this class I can...

1	**Parametric idea:** explain **parameters, constraints and design intent**	**Proof:** show which numbers drive our model
2	**Requirements:** turn the object to grip into **measured requirements and parameters**	**Proof:** a parameter table with units and limits
3	**Model:** build a **fully constrained** gripper finger driven by variables in Onshape (or OpenSCAD)	**Proof:** changing one variable resizes the whole part correctly
4	**Variants:** generate **two variants** for different objects by changing parameters only	**Proof:** two STL files from one model
5	**Test & iterate:** print, test the grip and **improve** one parameter	**Proof:** a test log and our version 2 value

LEARN

Objective 1 · Parametric idea · video

Video time!



What is Parametric & Non-Parametric Design?

Boldform Consulting · about 5 minutes · youtu.be/ERuOe8eLHOE

Watch for...

1. What is the difference between **parametric** and **direct** (non-parametric) modelling?
2. What is a **constraint**?
3. When might non-parametric modelling be better?

Scan to watch

After the video: Name an object in the room that comes in many sizes. Which 2 or 3 numbers would you make parameters?

LEARN

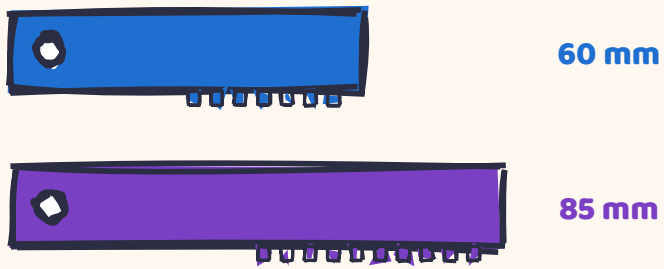
Objective 1 · Parametric idea

Parameters + constraints = design intent

The idea

Parameter: a named number that drives the model (**jaw_length = 60**).
Constraint: a rule that must stay true (hole is **centred**, edges are **parallel**, ridge spacing is **equal**).
Design intent: what should happen when a number changes. Make the finger longer: the hole stays near the pivot, the ridges spread along the tip.

Example: a recipe that scales

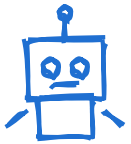


A cake recipe for 4 people uses 2 eggs and 200 g flour. For 8 people you double everything. The ratios (constraints) stay the same; one number (people) drives the rest. That is a parametric recipe!

LEARN

Real world

Parametric design in industry



Robot tooling
Grippers resized for each new product on the line



Cars
One part family, many models



Architecture
Stadium roofs and towers generated from parameters



Medical
Prosthetic hands sized from a patient's measurements

? e-NABLE volunteers 3D print free prosthetic hands. Why is parametric design perfect for that?

Customisation
Measure the person, type the numbers, print a hand that fits.

LEARN

Objective 4 · Variants

Variants: one model, many parts

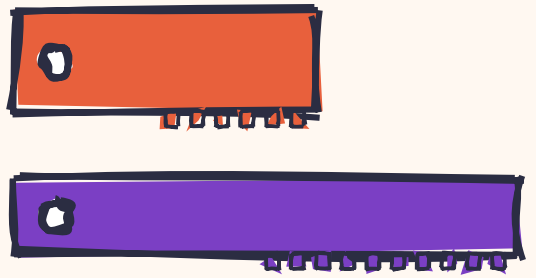
The idea

Make two versions **without redrawing anything**:

- **Egg**: object_width 45, ridge_depth 0.8 (gentle), jaw_width 14 (spread the force)
- **Bottle**: object_width 70, jaw_length 80, ridge_depth 2 (more grip)

In Onshape use a **Configuration** so both live in one part. Export each as its own STL.

Example: T-shirt sizes



egg

bottle

A shirt pattern is drawn once, then 'graded' to S, M, L and XL by changing a few measurements. Shops don't redesign the shirt for each size.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human CAD
2. From object to parameters
3. Onshape: variables and a sketch
4. Extrude, pattern, fillet
5. The same finger as code (OpenSCAD)
6. Print and assemble
7. Grip test

MAKE

Class game

Human CAD

How to play

1. Team A writes instructions to draw a gripper finger on squared paper, using ONLY parameters: L, W, hole_d.
2. Example: 'rectangle L by W; hole of diameter hole_d, centred, 8 from the left end'.
3. Team B draws it with L=10, W=3. Then again with L=16, W=4.
4. Does the second drawing still look right?

Then discuss

Where did the instructions break when the numbers changed? What constraint was missing?



Big idea

If the design breaks when a number changes, a constraint is missing.

Objective 2 · Requirements · plan it

From object to parameters

Parameter	Value	Unit	Why / limit
object_width	45	mm	width of the egg or bottle
jaw_length	60	mm	reach past the object's middle
jaw_width	12	mm	≥ 10 for strength in PLA
thickness	5	mm	matches servo horn
pivot_d	3.2	mm	M3 screw + 0.2 clearance
ridge_depth	1.5	mm	grip without crushing

Measure first

Use calipers on the real object and on the **SG90 servo** (body about 22.5 × 12.2 mm). Never trust an online drawing alone.

Units!

Write mm on everything. A gripper 45 cm wide is a famous classroom mistake.

Objective 3 · Build step 1 of 3

Onshape: variables and a sketch

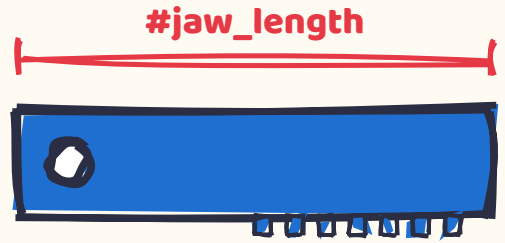
a Sign in to **Onshape** (free education account via the teacher). New document → Part Studio.

b Add a **Variable** feature for each parameter: **#jaw_length = 60 mm**, **#jaw_width = 12 mm** …

c Sketch on the Top plane: a rectangle and a circle for the pivot.

d Dimension with variables: type **#jaw_length** instead of 60.

e Add constraints until the sketch turns **black** (fully defined).



Check: Every line is black and every size is a variable.

Blue lines = not constrained. They will move unexpectedly when you change a number.

MAKE

Objective 3 · Build step 2 of 3

Extrude, pattern, fillet

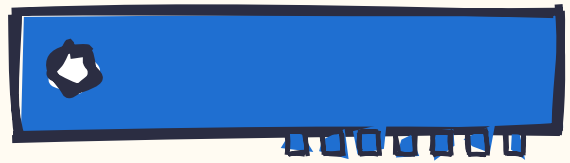
a **Extrude** the sketch by **#thickness**.

b Sketch one ridge on the gripping edge, extrude it by **#ridge_depth**.

c **Linear pattern** the ridge: count = **floor(#jaw_length / 8)**, so longer jaws get more ridges.

d **Fillet** outer corners 1 mm: stronger and kinder to fingers.

e Right-click the part → **Export** → STL.



Check: The finger has evenly spaced ridges along the tip and a clean pivot hole.

MAKE

Objective 3 · Model · code version

The same finger as code (OpenSCAD)

```
// Parametric gripper finger (OpenSCAD)
jaw_length  = 60;     // mm
jaw_width   = 12;
thickness   = 5;
ridge_depth = 1.5;
pivot_d     = 3.2;    // M3 + clearance

difference() {
  union() {
    cube([jaw_length, jaw_width, thickness]);
    for (x = [jaw_length * 0.5 : 4 : jaw_length - 4])
      translate([x, -ridge_depth, 0])
        cube([2, ridge_depth, thickness]);
  }
  translate([6, jaw_width / 2, -1])
    cylinder(d = pivot_d, h = thickness + 2, $fn = 40);
}
```

Code = CAD

OpenSCAD (free) builds 3D models from code. The parameters are just variables at the top.

Customizer

Window → Customizer shows the variables as sliders. Change them and watch the model update.

Why both?

Onshape is how industry works. OpenSCAD shows that a model can be a program. Try either!

MAKE

Objective 5 · Build step 3 of 3

Print and assemble

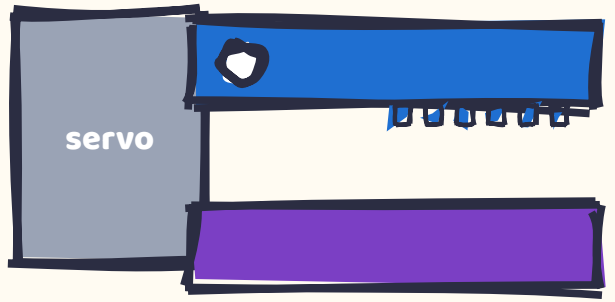
a Slice in Cura: PLA, 0.2 mm layers, **40% infill** (grippers need strength), print **flat**.

b Print one finger first as a test. About 20 minutes.

c Check the pivot hole with an M3 screw. Too tight? Change **pivot_d**, don't drill!

d Fix the finger to the servo horn with the small screw.

e Mount the second finger on the fixed side.



Check: The finger fits the servo horn and the pivot screw turns freely.

Hole too tight is a parameter problem. Fix it in the model so every future print is right.

MAKE

Objective 5 · Test & iterate · test it

Grip test

```
#include <Servo.h>
Servo grip;
int OPEN = 100, CLOSE = 40;    // tune for your gripper

void setup() { grip.attach(9); grip.write(OPEN); delay(1000); }

void loop() {
  grip.write(CLOSE); delay(2000);   // grip
  grip.write(OPEN);  delay(2000);   // release
}
```

Object	Holds?	Slips?	Damage?
Egg (or ping-pong ball)			
Bottle			
Pencil			
Shake test (5 s)			

Don't stall
If the servo buzzes while holding, CLOSE is too tight: it will overheat. Open by 5°.

MAKE

Objective 5 · Test & iterate · iterate

Version 2: change ONE parameter

Problem found	Parameter	Change
Egg slipped	ridge_depth	0.8 → 1.2
Bottle slipped	jaw_length	80 → 90
Finger bent	thickness	5 → 6
Pivot too tight	pivot_d	3.2 → 3.4

Engineering habit

Change one thing, re-test, record. Then the next. If you change 3 things at once you don't know which one helped.

Test like an engineer

Design log

Item	Result	Notes
Parameters defined (count)		
Sketch fully constrained?		
Variant A / B exported		
Print time and grams		
Grip test score		/ 4
V2 change and result		

Debug checklist

Model breaks when you change a number? A dimension or constraint is missing: look for blue lines.
Pattern count error? Use floor() so it is a whole number.

Good parametric models survive extreme values. Try jaw_length = 30 and 120. Does anything break?

MAKE

Level up!

Design challenges

Choose your challenge

- **Geared jaws:** add two spur gears (Onshape Spur Gear feature) so both jaws move together.
- **Fin-ray finger:** research the fish-fin finger that bends around objects.
- **Soft pads:** design slots for rubber-band pads. Better grip on smooth bottles?
- **Fit the arm:** make the mount fit last term's robot arm with parameters for its holes.

Think like an engineer

Why do soft grippers handle fruit better than hard metal ones?

Boss level

Make ONE model with a configuration for egg, bottle and pencil, and prove all three work.

Think about it

Design for people, not just objects

Scenario

Your parametric gripper could become a **prosthetic hand** for a child, sized from three measurements. Free files mean anyone can print one.

Discuss

- Who checks it is strong and safe enough?
- What if a print fails while someone is using it?
- Should medical designs be open for anyone to change?

Real example

e-NABLE volunteers worldwide have printed thousands of free hands, with careful testing and fitting guides.

Objective check

Can I really do it? Prove it.

01	What is the difference between a parameter and a constraint? Give one of each from your model.	○ ○ ○
02	Show your parameter table. Which value came from measuring the real object?	○ ○ ○
03	Change jaw_length live. Does everything update correctly?	○ ○ ○
04	Show your two variants. What changed between them?	○ ○ ○
05	Read your test log and your version 2 change.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Design review: teams swap models and try to 'break' them with extreme parameter values.

WRAP UP

Show what you know

What did we learn?

- [x] Parameters are named numbers that drive a model
- [x] Constraints keep the design intent when numbers change
- [x] Fully constrained sketches update predictably
- [x] Variants come from parameters, not redrawing
- [x] Print, test, change one parameter, repeat

At home

Measure three cups or bottles at home. Which parameter values would your gripper need for each?



Exit ticket: your pivot hole is too tight. What do you change, and why not just drill it bigger?

Next class: Motion-Controlled Game!